Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_19

1. Summary of Key Deviations

The analysis of Run_19 shows a major, system-wide loss of flow and pressure, indicating a severe physical fault.

- **Flow_Pump (Major):** The most significant anomaly is a major **negative offset**, with pump flow dropping by a substantial **-22.4%** compared to a healthy run. This points to a primary failure at or before the pump.
- **Flow_Bypass (Major):** This signal shows an even more severe **negative offset**, dropping by as much as **-44.6%**. This confirms that the entire system is starved of flow.
- **System-Wide Pressure Loss (Minor):** All pressure sensors (`p_Pump`, `p_Combustion`, `p_FMU`, `p_Shut`) show a consistent, minor **negative offset** of approximately **-5% to -6%**.
- **Flow_PRV & Flow_Engine (Nominal):** Crucially, both the Pressure Relief Valve flow and all four Engine flows show **no significant offset or fault**. `Flow_PRV` has a negligible drift, and `Flow_Engine` signals only show minor, non-problematic oscillations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Main Fuel Pump**	**Major Internal Leakage / Severe Degradation**	This is the most likely cause. A massive drop in **Flow_Pump** (-22.4%) is the "smoking gun." A severely degraded or internally leaking pump cannot generate the required flow, which perfectly explains the corresponding major drop in **Flow_Bypass** (-44.6%) and the minor, but system-wide, pressure loss.
2	**Boost-Pressurised Tank / Inlet Line**	**Severe Blockage / Clogged Filter**	A major blockage *before* the pump would starve it of fuel. This would produce symptoms nearly identical to a failed pump: a primary drop in `Flow_Pump` and subsequent drops in `Flow_Bypass` and all system pressures.

3	**Flow_Pump Sensor**	**Sensor Dropout / Major Negative Offset**	It's possible the `Flow_Pump` sensor is faulty and reading much lower than reality. However, this is less likely because all the other pressure and flow signals are physically consistent with a real loss of flow. A sensor-only fault would not cause every other component's reading to drop in a physically plausible way.
4	**Bypass Valve**	**Stuck Open**	If the bypass valve were stuck significantly open, it would divert flow away from the main system. However, in this scenario, the **Flow_Bypass** sensor should read *high* (as it measures the diverted flow), but it is reading very low. This makes a stuck-open bypass valve an unlikely root cause.
5	**Major Leak (Post-Pump)**	**Leak in Piping Before FMU**	A major leak in the pipes between the pump and the FMU could cause a drop in measured flow and pressure. However, the drop in `Flow_Pump` is so significant that it points more towards the pump itself being the source of the problem rather than a downstream leak.

3. Most Probable Hypothesis

The most probable root cause is a **Major Internal Leakage or Severe Degradation of the Main Fuel Pump**.

Reasoning: The evidence overwhelmingly points to a catastrophic failure of the main pump. The primary symptom is the massive -22.4% drop in measured flow directly from the pump. This single upstream failure is sufficient to explain every other observed physical symptom: the system is starved of fuel, causing the bypass flow to drop dramatically and all downstream pressures to fall. The fact that the engine and PRV flows remain nominal (relative to the available flow) further isolates the problem to the pump itself or a blockage immediately before it. Given the severity, a degraded pump is the most direct and comprehensive explanation.